Formulation of Signed Area — Derivation of Charge Structure and Consequences of Spin-2 Configurations

— Three Consequences Derived from the Sign Product P of [4] §7 —

Author: Noriaki Kihara (WF System Co., Ltd.)

ORCID: 0009-0006-7814-2793

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§0 Purpose of This Paper

In the remark of [4] §7, it was stated that "the orientation structure of the signed area $M(\sigma)$ for square faces containing the R-axis, and the rigorous definition of conserved quantities based thereon, lie beyond the scope of this paper and will be addressed in a separate work."

This paper responds to that declaration and derives the following three results.

1. The rigorous definition of the signed area M and its relation to the sign product P of [4] Proposition 7.1 (§1)
2. The definition of the fermion area M_F and its properties concerning antiparticles and color charge (§2)
3. Consequences of the $P = -1$ configuration in spin-2 tR-type (§3)

§1–§3 are self-contained within the combinatorial framework of [4]. §4 adopts the same stance as [4] §9 as an interpretive example, showing the correspondence between M_F and charge quantization.

Referenced papers: - [3] Full Spherical Coverage of Central Projection in Discrete Space and the Number-Theoretic Necessity of Five-Dimensional Background Space (W5) - [4] Set Structure of the Six-Dimensional Hyperrectangle and Its Combinatorial Properties (W8) - [5] Interactions of Shape-Invariant Waves (W11)

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§1 Definition of Signed Area

1.1 2-Faces

We use the set signs from [4] §1.3.3.

Definition 1.1 (2-face). Among the six axes (x, y, z, t, R, Q), a set in which exactly two axes i, j have nonzero signs ($\sigma_i, \sigma_j \in \{+1, -1\}$) and all remaining axes have zero signs is called the **2-face** formed by axes (i, j).

1.2 Signed Area

By [4] §1.1, each coordinate can take nonzero values $v_i, v_j \neq 0$ (either real-valued or integer-valued).

Definition 1.2 (Signed area). For the 2-face formed by axes (i, j), the **signed area** $M(i, j)$ is defined as:

$$M(i,j) = v_i \cdot v_j$$

M is real-valued and carries two pieces of information: its **sign** (orientation) and its **absolute value** (magnitude of area):

$$\text{sign}(M) = \text{sign}(v_i) \cdot \text{sign}(v_j) \in \{+1, -1\}$$

$$|M| = |v_i| \cdot |v_j|$$

$\text{sign}(M) = +1$ is called **positive orientation** and $\text{sign}(M) = -1$ is called **reverse orientation**. The specific values of $|v_i|$ and $|v_j|$ are not constrained by this definition.

Remark. For cancellation of signed areas (e.g., particle-antiparticle pair annihilation) to satisfy $M + M' = 0$, the condition $|M| = |M'|$, i.e., equality of the absolute values of the axes, is required. The antiparticle definition in [5] §5.4 (only spatial axis signs are inverted; absolute values are invariant) automatically satisfies this condition.

Proposition 1.2 (Signed area on integer lattices). By [3], coordinates that realize full spherical coverage of central projection in discrete space take values on the integer lattice $\mathbb{Z}^n$. Then for nonzero coordinates $v_i, v_j \in \mathbb{Z} \setminus \{0\}$:

(i) $M(i,j) = v_i \cdot v_j \in \mathbb{Z} \setminus \{0\}$

(ii) $\text{sign}(M) \in \{+1, -1\}$

Proof. (i) The integer ring $\mathbb{Z}$ is closed under multiplication ($a, b \in \mathbb{Z} \implies ab \in \mathbb{Z}$) and is an integral domain ($a \neq 0$ and $b \neq 0 \implies ab \neq 0$). Since $v_i, v_j \in \mathbb{Z} \setminus \{0\}$, we have $M = v_i \cdot v_j \in \mathbb{Z} \setminus \{0\}$. (ii) Since $M \neq 0$, $\text{sign}(M) \in \{+1, -1\}$ is well-defined. □

Remark. Definition 1.2 is also applicable to the continuous case where coordinate values are real-valued, but the two-valuedness of $\text{sign}(M)$ is guaranteed solely by the condition $v_i, v_j \neq 0$. On integer lattices, this condition is automatically satisfied by the lattice structure.

1.3 Relation to the Sign Product P

Proposition 1.1. The sign product $P = (-1)^{|\{i \in \{t,R\} : \sigma_i < 0\}|}$ from [4] Proposition 7.1 coincides with the sign of $M(t,R)$ when both the t-axis and R-axis are nonzero: $P = \text{sign}(M(t,R))$.

Proof. When $\text{sign}(v_t), \text{sign}(v_R) \in \{+1, -1\}$:

$\text{sign}(v_t)$	$\text{sign}(v_R)$	Number of negatives	P	$\text{sign}(M)$
$+$	$+$	0	$+1$	$+1$
$+$	$-$	1	-1	-1
$-$	$+$	1	-1	-1
$-$	$-$	2	$+1$	$+1$

$P = \text{sign}(M)$ holds for all four configurations. □

Remark. When only one of t, R is nonzero, $P = (-1)^{0 \text{ or } 1}$, and the 2-face $M(t,R)$ is not defined (a single axis does not constitute a face). In this case, P functions as a generalization of $\text{sign}(M)$.

§2 Fermion Area

2.1 Definition

By [4] Definition 2.1, fermion types have exactly $k = 2$ of the four $xyzt$ axes nonzero.

Definition 2.1 (Fermion area). The signed area $M_F = v_i \cdot v_j$ of the 2-face formed by the two nonzero axes $(i, j) \subseteq \{x, y, z, t\}$ of a fermion is called the **fermion area**. Its **orientation** is given by $\text{sign}(M_F) = \text{sign}(v_i) \cdot \text{sign}(v_j) \in \{+1, -1\}$.

2.2 Orientation of Fermion Area in [4] Table A

We compute $\text{sign}(M_F)$ for all fermions in [4] §9.9 Table A. The absolute values $|v_i|$ of each axis do not affect the results of this section.

Charged leptons (nonzero axes: 1 spatial axis + t-axis, $v_t > 0$, $Q = 4$, $c_1 = 1$):

Particle	Set	Nonzero axes	$\text{sign}(M_F)$
e^-	$(+, 0, 0, +, 0, 4)$	x, t	$(+)(+) = +1$
μ^-	$(0, +, 0, +, 0, 4)$	y, t	$(+)(+) = +1$
τ^-	$(0, 0, +, +, 0, 4)$	z, t	$(+)(+) = +1$

Neutrinos (nonzero axes: 2 spatial axes, $t = 0$):

Particle	Set	Nonzero axes	$\text{sign}(M_F)$
ν_e	$(+, +, 0, 0, 0, 0)$	x, y	$(+)(+) = +1$
ν_μ	$(0, +, +, 0, 0, 0)$	y, z	$(+)(+) = +1$
ν_τ	$(+, 0, +, 0, 0, 0)$	x, z	$(+)(+) = +1$

Up-type quarks (nonzero axes: 1 spatial axis + t-axis, $v_t > 0$, $Q \in \{5, 6, 7\}$, $c_1 = 1$):

Particle	Set	Nonzero axes	$\text{sign}(M_F)$
u	$(+, 0, 0, +, 0, Q)$	x, t	$(+)(+) = +1$
c	$(0, +, 0, +, 0, Q)$	y, t	$(+)(+) = +1$
t	$(0, 0, +, +, 0, Q)$	z, t	$(+)(+) = +1$

Down-type quarks (nonzero axes: 1 spatial axis + t-axis, $v_t < 0$, $Q \in \{1, 2, 3\}$, $c_1 = 0$):

Particle	Set	Nonzero axes	$\text{sign}(M_F)$
d	$(+, 0, 0, -, 0, Q)$	x, t	$(+)(-) = -1$
s	$(0, +, 0, -, 0, Q)$	y, t	$(+)(-) = -1$
b	$(0, 0, +, -, 0, Q)$	z, t	$(+)(-) = -1$

Summary: It is solely the **orientation** (sign) of M_F that determines the charge structure; it does not depend on the absolute value $|M_F| = |v_i| \cdot |v_j|$.

Fermion type	Type of nonzero axes	Q	$\mathrm{sign}(M_F)$
Charged lepton	(spatial, $t > 0$)	4	$+1$
Neutrino	(spatial, spatial)	0	$+1$
Up-type quark	(spatial, $t > 0$)	5,6,7	$+1$
Down-type quark	(spatial, $t < 0$)	1,2,3	-1

2.3 Fermion Area of Antiparticles

By [5] §5.4, for antiparticles the signs of the spatial axes (x, y, z) are inverted, while the sign of the t-axis and the Q value remain unchanged.

Proposition 2.1. When the fermion face consists of a (spatial, t) combination, the fermion area of the antiparticle has the opposite sign to that of the particle: $M_F^{\text{anti}} = -M_F$.

Proof. For the antiparticle, the spatial axis value is inverted to $-v_{\text{spatial}}$, while the t-axis remains unchanged. Therefore, $M_F^{\text{anti}} = (-v_{\text{spatial}}) \cdot v_t = -(v_{\text{spatial}} \cdot v_t) = -M_F$. □

Proposition 2.2. When the fermion face consists of a (spatial, spatial) combination, the fermion area of the antiparticle is identical to that of the particle: $M_F^{\text{anti}} = M_F$.

Proof. For the antiparticle, both spatial axis values are inverted to $-v_{\text{spatial}_1}, -v_{\text{spatial}_2}$. Therefore, $M_F^{\text{anti}} = (-v_{\text{spatial}_1}) \cdot (-v_{\text{spatial}_2}) = v_{\text{spatial}_1} \cdot v_{\text{spatial}_2} = M_F$. □

Corollary 2.1. Neutrinos in [4] Table A (spatial × spatial, $\mathrm{sign}(M_F) = +1$) have antineutrinos that also possess $\mathrm{sign}(M_F) = +1$. For charged leptons and all quarks (spatial × t), the antiparticle has M_F with inverted sign ($\mathrm{sign}(M_F^{\text{anti}}) = -\mathrm{sign}(M_F)$).

2.4 Independence of Color Charge and Fermion Area

Proposition 2.3. The value of the Q-axis does not affect the fermion area M_F. Fermions with the same $\sigma_{\text{spatial}}, \sigma_t$ have identical M_F regardless of the value of Q.

Proof. $\mathrm{sign}(M_F) = \mathrm{sign}(v_i) \cdot \mathrm{sign}(v_j)$ is determined solely by the values of the $xyzt$ axes, and Q does not appear in the definition. □

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§3 The $P = -1$ Configuration of Spin-2 tR-Type

By [4] Proposition 7.1, spin-2 tR-type admits both $P = +1$ and $P = -1$ configurations. In [4] §9.7, the physical correspondence of $P = -1$ was left unresolved.

Proposition 3.1. The $P = -1$ configuration of spin-2 tR-type is equivalent to σ_t and σ_R having opposite signs.

Proof. By Proposition 1.1, $P = \mathrm{sign}(M(t, R)) = \mathrm{sign}(v_t) \cdot \mathrm{sign}(v_R)$. $P = -1 \iff \mathrm{sign}(v_t) \cdot \mathrm{sign}(v_R) = -1 \iff v_t$ and v_R have opposite signs. □

Proposition 3.2. Applying the sign rule for attraction/repulsion from [5] §4.3 (same relative sign of t-axis components implies attraction; opposite signs imply repulsion) to the graviton interpretation of [4] §9.5:

- $P = +1$ (t, R same sign): t-component has a single sign → attraction only
- $P = -1$ (t, R opposite signs): t-component and R-component have opposite signs → repulsive along the R-axis direction

Proof. In [5] §4.3, the sign of interaction between direction-constrained waves is determined by $\mathrm{sign}(k_t^{(1)}) \cdot \mathrm{sign}(k_t^{(2)})$. When the spherical wave of spin-2 tR-type follows this sign rule, $P = +1$ implies both axes share the same directionality, yielding attraction only. For $P = -1$, the action along the R-direction has the opposite sign to the t-direction, acting repulsively with respect to the spatial scale. □

Remark. The interpretation of the physical consequences of the $P = -1$ configuration is addressed in §4.3.

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§4 Interpretive Example: Derivation of Charge Structure

This section has the same status as [4] §9; it is an interpretive example that maps the combinatorial results of §1–§3 onto the Standard Model.

4.1 Third Component of Weak Isospin I_3

Using the $\mathrm{sign}(M_F)$ introduced in §2.2 and the c_1 bit (weak isospin) from [4] §9.2, we define the third component of weak isospin I_3 of the Standard Model as follows.

For quarks ($c_2c_3 \neq 00$), since $\mathrm{sign}(M_F)$ directly distinguishes up-type ($+1$) from down-type (-1), we set $I_3 = \mathrm{sign}(M_F)/2$.

For leptons ($c_2c_3 = 00$), neutrinos ($Q = 0$, spatial $\times$ spatial) and charged leptons ($Q = 4$, spatial $\times$ t) both have $\mathrm{sign}(M_F) = +1$ and cannot be distinguished by $\mathrm{sign}(M_F)$ alone. We define $I_3 = (1 - 2c_1)/2$ using the c_1 bit.

Fermion type	Q	c_1	$\mathrm{sign}(M_F)$	I_3	SM I_3
Neutrino	0	0	$+1$	$(1 - 2 \cdot 0)/2 = +1/2$	$+1/2$
Charged lepton	4	1	$+1$	$(1 - 2 \cdot 1)/2 = -1/2$	$-1/2$
Up-type quark	$5, 6, 7$	1	$+1$	$+1/2$	$+1/2$
Down-type quark	$1, 2, 3$	0	-1	$-1/2$	$-1/2$

The value of I_3 agrees with the Standard Model for all fermions. For quarks, $\mathrm{sign}(M_F) = 2c_1 - 1$ holds, so $I_3 = \mathrm{sign}(M_F)/2 = (2c_1 - 1)/2$ has the opposite sign relation to $(1 - 2c_1)/2$ for leptons.

4.2 Determination of Weak Hypercharge Y from Color Charge Components c_2c_3

The Standard Model charge formula (Gell-Mann–Nishijima relation):

$$Q_{\text{charge}} = I_3 + \frac{Y}{2}$$

Assume that Y depends only on the color charge components c_2c_3: $Y = Y_L$ when $c_2c_3 = 00$ (leptons), and $Y = Y_Q$ when $c_2c_3 \neq 00$ (quarks). Then the following two conditions uniquely determine Y_L, Y_Q.

Condition (i): Integer charges for leptons. For any fermion with $c_2c_3 = 00$, $Q_{\text{charge}} = I_3 + Y_L/2 \in \mathbb{Z}$ must hold. Since $I_3 \in \{+1/2, -1/2\}$, Y_L must be odd.

Condition (ii): Integer charges for color-neutral baryons. The total charge of a three-body color-neutral system ($c_2c_3 = 01, 10, 11$, one each) with $c_2c_3 \neq 00$, $Q_{\text{baryon}} = \sum_{a=1}^{3}(I_3^{(a)} + Y_Q/2) \in \mathbb{Z}$, must hold for any combination of $I_3^{(a)} \in \{+1/2, -1/2\}$.

$$Q_{\text{baryon}} = \sum I_3^{(a)} + \frac{3Y_Q}{2}$$

Since $\sum I_3^{(a)} \in \{-3/2, -1/2, +1/2, +3/2\}$ (always a half-integer), $3Y_Q/2$ must also be a half-integer. That is, $Y_Q = (2n+1)/3$ ($n \in \mathbb{Z}$).

Determination of the minimal solution. From condition (i), the odd number with smallest $|Y_L|$ is $Y_L = \pm 1$. Requiring negative charge for charged leptons:

$$Q_{\text{charge}}(e^-) = -\frac{1}{2} + \frac{Y_L}{2} < 0 \implies Y_L < 1$$

Hence $Y_L = -1$. From condition (ii), the solution with smallest $|Y_Q|$ is $n = 0$: $Y_Q = 1/3$.

Proposition 4.1 (Unique determination of Y from integer charge conditions). From the two-valuedness of $I_3 \in \{+1/2, -1/2\}$ and the condition that charges of observable particles (leptons and color-neutral baryons) are integers, the minimal solution for Y is uniquely determined as:

$$Y_L = -1 \quad (c_2c_3 = 00), \qquad Y_Q = +\frac{1}{3} \quad (c_2c_3 \neq 00)$$

4.3 Consequences

Consequence 1: Reproduction of the Gell-Mann–Nishijima formula. Substituting the I_3 from §4.1 and Y from Proposition 4.1:

Fermion	sign(M_F)	I_3	Y	$Q_{\text{charge}} = I_3 + Y/2$	SM value
ν	$+1$	$+1/2$	-1	0	0
e^-	$+1$	$-1/2$	-1	-1	-1
u	$+1$	$+1/2$	$+1/3$	$+2/3$	$+2/3$
d	-1	$-1/2$	$+1/3$	$-1/3$	$-1/3$

This is consistent with the charge map f_Q^{charge} of [4] Definition 9.1.

Consequence 2: Automatic satisfaction of anomaly cancellation. The solution from Proposition 4.1 satisfies:

$$Y_L + 3Y_Q = -1 + 3 \times \frac{1}{3} = 0$$

This corresponds to the gauge anomaly cancellation condition in the Standard Model. In this framework, it is derived algebraically from the structure of c_2c_3 (one color-neutral state $c_2c_3 = 00$ + three color charge states $c_2c_3 \neq 00$) and the integer charge condition.

Consequence 3: Charge equality of proton and electron. The charge of the proton (uud):

$$Q_p = \sum I_3^{(a)} + \frac{3Y_Q}{2} = \left(\frac{1}{2} + \frac{1}{2} - \frac{1}{2}\right) + \frac{1}{2} = +1$$

The charge (absolute value) of the electron:

$$|Q_e| = \left|I_3(e^-) + \frac{Y_L}{2}\right| = \left|-\frac{1}{2} - \frac{1}{2}\right| = 1$$

$|Q_p| = |Q_e| = 1$; the reason that the proton and electron carry charges of equal magnitude is determined algebraically from the color charge structure of c_2c_3 ($1+3$ classification) and the two-valuedness of I_3 ($\pm 1/2$). There is no need to introduce fractional charges $+2/3, -1/3$ as fundamental quantities.

Consequence 4: Interpretation of spin-2 $P = -1$. By Proposition 3.2 of §3, the graviton configuration with $P = -1$ acts repulsively along the R-axis direction. Since the R-axis in [4] §9.8 is the axis that links the subjective coordinate system to the spatial scale of the six-dimensional structure, the $P = -1$ configuration may correspond to **accelerating expansion of the spatial scale**. This is a candidate for the geometric description of the cosmological constant $\Lambda > 0$ (dark energy). However, whether the $P = -1$ configuration is physically realized (existence of selection rules) remains an open question, and this paper limits itself to stating the consequence.

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§5 Conclusion

In this paper, we provided the definition of the signed area $M(\sigma)$ declared in [4] §7 and derived the following three results.

1. The **signed area** $M(i,j) = v_i \cdot v_j$ (Definition 1.2) coincides with the restriction of the sign product P of [4] Proposition 7.1 to the case where both t and R are nonzero (Proposition 1.1).
2. The **fermion area** M_F (Definition 2.1) is computable for all fermions in [4] Table A, and we demonstrated sign inversion under antiparticles (Proposition 2.1), invariance of neutrino antiparticles (Proposition 2.2), and independence from the Q-axis (Proposition 2.3).
3. The **spin-2** $P = -1$ configuration corresponds to opposite signs of v_t, v_R, and from the sign rule of [5], it was derived to be repulsive along the R-axis direction (Propositions 3.1, 3.2).

In the interpretive example (§4): - I_3 (quarks: $\mathrm{sign}(M_F)/2$; leptons: $(1-2c_1)/2$) agrees with the third component of weak isospin in the Standard Model (§4.1) - The integer charge condition uniquely determines $Y_L = -1, Y_Q = +1/3$ (Proposition 4.1) - The anomaly cancellation condition $Y_L + 3Y_Q = 0$ is automatically satisfied (Consequence 2) - The charge equality of the proton and electron is derived from the $1+3$ structure of c_2c_3 (Consequence 3)

were demonstrated.

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